

XINGJIAN MA

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Education

University of Connecticut

Sep 2024 – May 2028 (Expected)

PhD Candidate in Statistics, Department of Statistics

Storrs, CT, U.S.

- Courses: Design of Experiments, Statistical Inference, Linear Models

Renmin University of China

Sep 2022 – Jun 2024

Master of Science in Applied Statistics, Institute of Statistics and Big Data

Beijing, China

- Courses: Clinical Trial Statistical Methodology, Causal Inference, Survival Analysis

Experience

Astellas Pharma

May 2026 – Aug 2026

Biostatistician Intern

Remote

Closed-Form Evaluation of Correlated PFS and OS Group Sequential Designs

- Developed an event-driven group sequential design for correlated progression-free survival (PFS) and overall survival (OS) under the Fleischer model, supporting efficacy monitoring, futility stopping, and piecewise uniform accrual.
- Derived a closed-form joint correlation matrix for sequential PFS and OS log-rank statistics, reported stopping boundaries on z-score, p-value, and hazard-ratio scales, calculated marginal power and gatekeeping joint power and validated the results against Monte Carlo simulations ($< 10^{-4}$ absolute discrepancy).
- Implemented the methodology in an R package and Shiny application, deployed via Posit Connect and used by Astellas statisticians for trial-design comparisons; presented the work at an internal statistics meeting and incorporated user feedback to improve the tools' functionality and usability.

Department of Public Health Sciences, UConn Health

Sep 2025 – Dec 2025

Research Assistant

Farmington, CT, U.S.

Longitudinal Analysis of Depression Outcomes in Large-Scale Medicare Claims Data

- Processed and linked over 3 billion Medicare claims records using `fst` and `data.table` to construct a longitudinal cohort of more than 2 million beneficiaries with depression, integrating claims history, chronic-condition indicators, and demographic covariates.
- Used Cox proportional hazards models with annual follow-up intervals to estimate associations of clinical and demographic factors with time to depression remission, defined as the first annual assessment at which a beneficiary no longer met the study's depression criterion based on the depression score.

Center for Population Health, UConn Health

Jan 2025 – Jun 2025

Research Assistant

Farmington, CT, U.S.

Survival Analysis of Suicide Risk Using Clinical Data

- Analyzed over 5 million longitudinal clinical encounter records and engineered predictors from ICD-9/10 diagnosis histories, Charlson comorbidity scores, medication exposures, and demographic characteristics.
- Used Weibull and Cox proportional hazards models to identify clinical and demographic factors associated with time to the first suicide-related event; used logistic regression of the binary suicide indicator as a complementary analysis to assess the consistency of identified risk factors.

Projects

CorrSurvGSD | R, Shiny

2026

- Built an R package with Shiny app that enables statisticians to specify two-arm sample sizes and median survival times, accrual rates, and group sequential monitoring settings, including efficacy and futility rules, for correlated PFS/OS designs; generates calendar cutoffs, stopping boundaries, marginal power, and gatekeeping joint power.

rerand | R, Rcpp

2025

- Developed an R package for rerandomization experiments, allowing users to select covariates separately for treatment assignment and analysis; provides difference-in-means, ANCOVA, and Lin estimators with conservative variance estimates, and Gaussian-mixture-based confidence intervals and hypothesis tests.
- Implemented efficient Rcpp routines for large-scale rerandomization, achieving a 5x speedup in generating treatment assignments over pure R.

Skills

Biostatistics: Clinical Trial Design (Group Sequential, Sample Size Re-estimation, Adaptive Design), Survival Analysis (Cox, Parametric, Competing Risks), Longitudinal Data Analysis (Mixed Models, GEE), Monte Carlo Simulation

Programming: R (package development, `data.table`, Shiny, `ggplot2`), SAS, Rcpp, Python (NumPy, pandas), SQL

Tools: Git/GitHub, Posit Connect, Linux/Bash, HPC (Slurm), Markdown/LaTeX